

THE LAST SEED

Behind the scenes

How a 60-second pixel-art film was made entirely in code — by a team of 14 AI agents

- 60 seconds • 1080p • 30 fps
- 0 AI video generators
- 14 agents
- Original score & sound design

Directed by Avitz Meir (Avlz - AI for Humans) • Produced with Claude

01

The brief

One sentence, no constraints, a very high bar

> **Create a beautiful pixel-art animation with a great story. 1-minute video. Use up to 15 agents. BLOW ME AWAY — make it so impressive that no one could imagine it was made by an AI and not a professional team in months.**

The decision that shaped everything

Pixel art made by an image or video model always gives itself away: mixed pixel sizes, smeared edges, drifting palettes. So no generator was used at all. Every pixel, every frame of animation, every note and every sound was written as code — the way a small game studio would build it, but staffed by AI agents.

In numbers

60 s
film length

1,800
frames, each a pure function of time

320×180
native pixels, shown 6× at 1080p

132
palette colors, in hand-tuned ramps

20
shots incl. 8 close-up inserts

14
AI agents (11 in parallel)

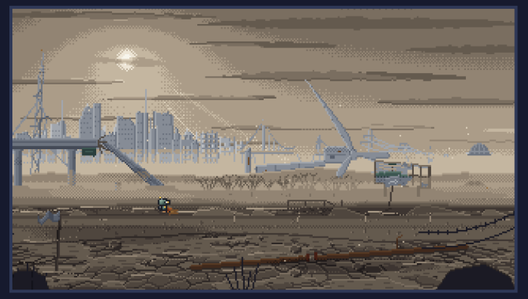
21
issues found by the critic — all fixed

0
samples, stock assets or video models

02

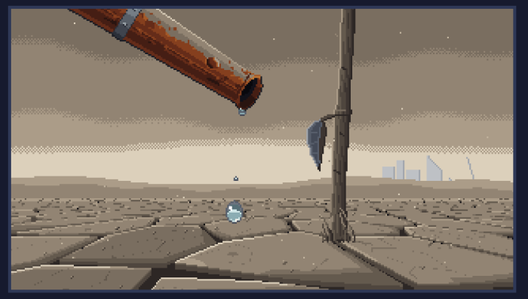
The story

Wordless. Sixty seconds. One idea: giving your light away is how the world gets its color back.



1. A dead world

A small rusty gardening robot crosses a grey wasteland. Its chest core is the only warm light.



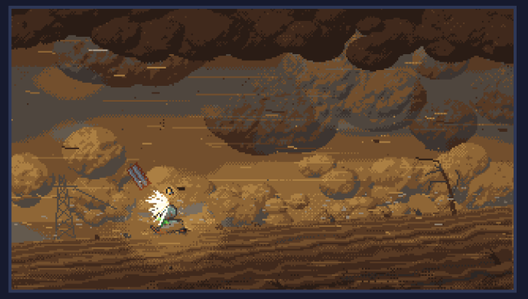
2. The empty can

It waters a dead plant out of habit. One drop falls. The earth swallows it.



3. The last seed

A gust uncovers a sprout — the first green in the film. It carries it home in its can.



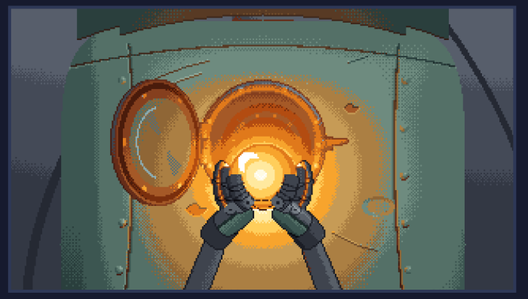
4. The storm

A sandstorm tears off its antenna. It curls around the sprout to protect it.



5. The realization

At night the dying sprout leans toward the glow of the robot's chest.



6. The gift

At dawn it opens its chest and plants its own heart beside the sprout.



7. Rebirth

A tree bursts through the greenhouse; a wave of color rolls across the land.



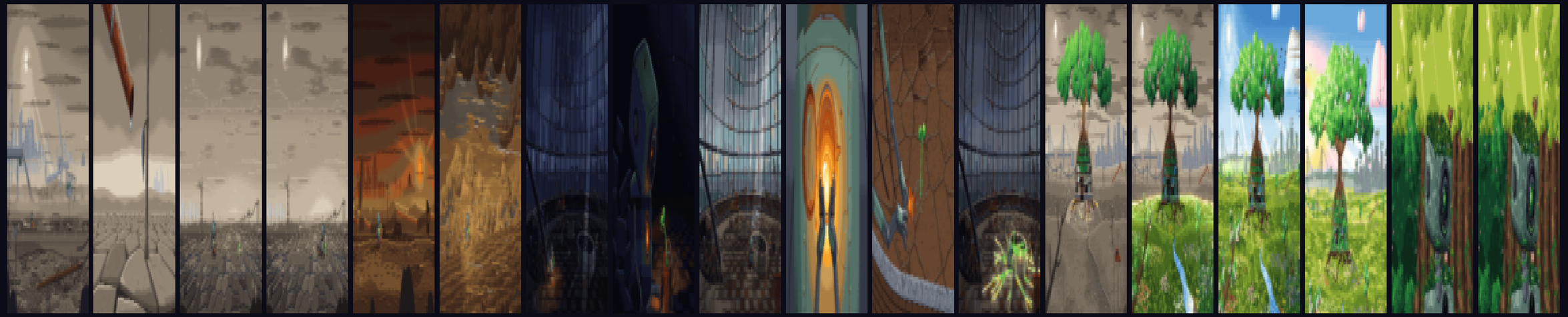
8. Rest

Seasons later, a bird lands on the mossy robot — and its eye glows green.

03

Color is the story

The first half has almost no saturated color. The robot's amber heart and the sprout's green are the only exceptions.



2s 9s 13.6s 18s 23s 27.5s 33s 36s 39s 41.5s 43.5s 46.5s 48.5s 50s 51.5s 54.5s 56.5s 57.5s

How it is enforced

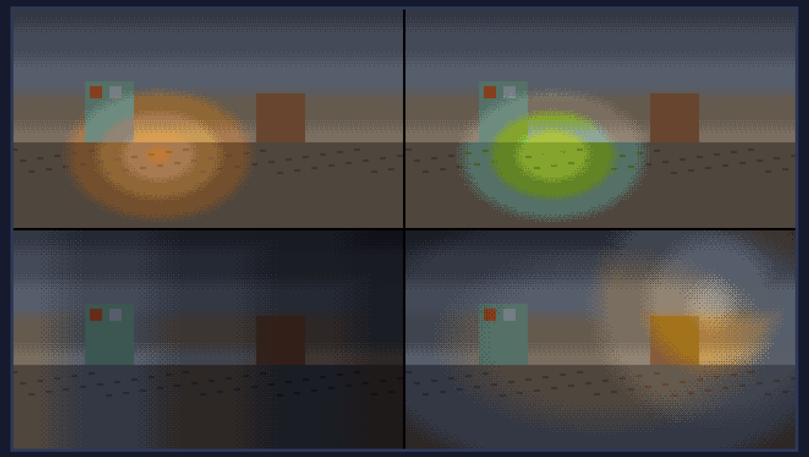
- A single master palette of 132 colors organised as dark-to-light ramps with hue-shifted shadows. Every frame is snapped to it.
- "Dead world" ramps (dust, ash, ochre, rust, steel, faded mint) for shots 1–7. The leaf ramp is reserved for the sprout alone.
- When the robot gives its heart, the bloom shot composites a dead and an alive version of the same world behind an expanding wavefront.

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The engine

Built first, by the director, so 13 artists could work in parallel without breaking the look





Palette-true lighting: warm, life, shadow and sun light pools

The contract every shot obeys

```
from engine import *
DURATION = 6.0
def render(t):
    ...   # palette colors only
    return frame   # 320x180 or Stage
```

Rules of the engine

- 320×180 canvas, upscaled 6× with nearest-neighbour: one pixel size for the whole film.
- render(t) is a pure function of time — any frame can be rendered alone, out of order, in parallel.
- Light is measured in palette steps and dithered at the edges, so lit pixels stay inside the palette.
- A layered Stage composites parallax at sub-pixel positions, so slow pans glide instead of judder.
- Fades walk every color down its own ramp: no hue shifts, no screen-door dither.

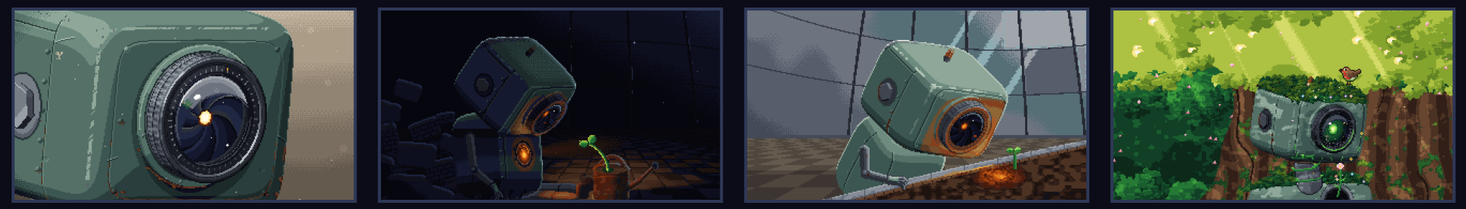
05

One robot, drawn by many hands

A canonical 26-pixel model sheet that every animator and illustrator had to match



Model sheet: normal and post-storm "damaged" variant



The same robot as large close-up illustrations (eye, realization, last light, rebirth)

Design language

- Big head, one round camera-lens eye with an amber pupil, a porthole on the chest that shows its glowing core.
- Faded mint paint, rust only on seams and lower edges, a hex ear-bolt, a cracked solar backpack.
- A continuity table tracks antenna, damage, core brightness, eye and the can's contents in every shot.

Animated as code

- Every animation frame is hand-authored ASCII pixel art, assembled like a paper-doll rig.
- A shared API: robot.frame(anim, t, can, variant, core, eye) with anchors for feet, eye, core and hands.
- Hand-off constants (e.g. the exact moment the soil clump leaves the ground) keep props and actors in sync across agents.

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The team

14 AI agents, each owning its files. 11 worked in parallel in the first phase.

- **Director**
  Story, engine, palette, bible, compositor, mix, edit
- **Character animator A**
  Walk, water, kneel, scoop, storm acting, props
- **Character animator B**
  Sitting, digging, opening its chest, powering down
- **Wasteland artist**
  S1 dead world, S4 journey at dusk
- **Discovery artist**
  S2 empty can, S3 discovery
- **Close-up illustrator**
  Eye, sprout, heart, rest; later the lean and the last light
- **Storm FX artist**
  S5 sandstorm, haboob, antenna strike
- **Greenhouse lighting artist**
  S6 night, S7 the gift, S8a bloom inside
- **Bloom FX artist**
  S8b the climax, S9a the valley
- **Title designer**
  Custom pixel font and animated end title
- **Composer**
  Fully synthesized score, two themes
- **Sound designer**
  Ambience, foley, robot voice, synced to frames
- **Inserts artist**
  The last drop, the heart into the soil
- **Critic**
  Reviewed all 1,800 frames and the audio; 21 issues
- **Close-up polish artist**
  Unified the robot across close-ups, the flare, the bird

07

The pipeline

Build the rules first, then parallelise, then review ruthlessly

1. Foundations

Engine, palette, model sheet, production bible, cue sheet, stub shots — the whole film renders end-to-end on day zero.

2. Parallel production

11 agents build characters, 6 shot groups, score, sound and title against the same contract. Each renders and looks at its own frames, 4+ critique cycles.

3. First cut

Director assembles, mixes and reviews contact sheets. Verdict: beautiful, but the robot is too small at the emotional moments.

4. Close-up inserts

Five new close-ups carry the key beats: the last drop, the lean, the heart, the last light.

5. Fresh-eyes critic

An independent agent audits every frame, cut and sync point and ranks 21 issues by severity.

6. Fix round

Each issue goes back to its owner. Audio re-renders automatically from the cue sheet.

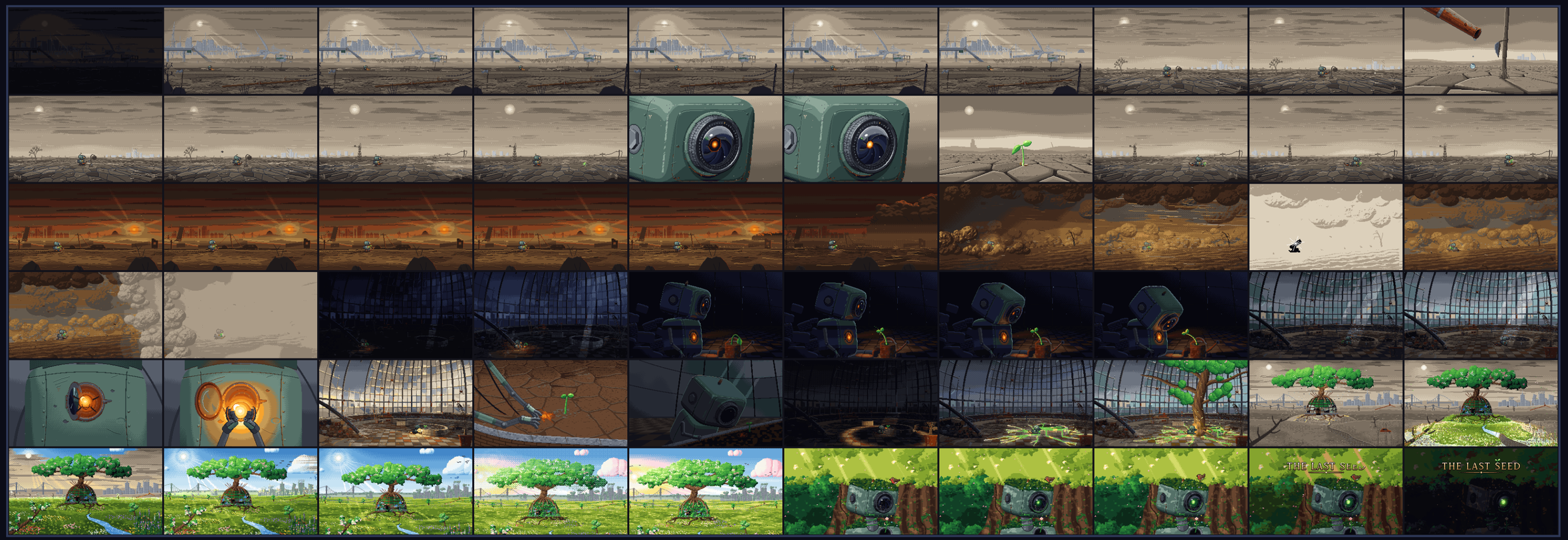
7. Master

Frame-exact fades, final mix at −15.6 LUFS, 1080p H.264.

08

The edit

Twenty shots on a 60-second clock; medium shots keep running "off-screen" under their close-ups



One frame per second of the final film. Every sync point lives in one file (cues.py); pictures, music and sound are all built from it, so moving a cue moves everything with it.

09

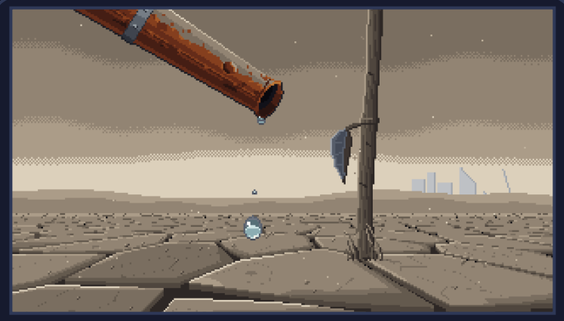
Review round one: the close-ups

In the first cut the robot was 26 pixels tall in a 180-pixel frame — at the very moments that mattered most



Before: the beat is there, but nobody can see it

Fix: close-up inserts drawn as large illustrations at the same pixel scale. The medium shot's clock keeps running underneath, so cutting back is seamless. Three eye close-ups now rhyme across the film: amber when it finds the sprout, dying when it gives its heart, green when it wakes.



The last drop



The lean



The heart into the soil



The last light

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The critic

A separate agent with fresh eyes, told to be ruthless — 21 findings, every one fixed by its owner

Must-fix

- 9.600 s: the drop's impact frame was blank — float rounding (9.6 − 8.9 = 0.6999…). Fixed: frame-snapped clocks everywhere.
- 43.4 s: the light jumped from dawn to night mid-sacrifice. Fixed: the close-up opens at dawn exposure, then darkens.
- Storm turn: the robot popped to full height for 3 frames and the sprout vanished. Fixed: a crouched front view with the can.
- S4: the hero vibrated 4 px at 10 Hz from sub-pixel phase locking. Fixed: robot on its own integer layer.

Should-fix

- The heart going into the soil was barely visible — new close-up insert.
- The lean read as "perks up", not "reaches for the light" — visible light path, cause and effect with the flicker.
- Close-ups drifted off-model (three ear-bolts, three hand designs) — one canonical close-up robot.
- Procedural tells: glyph birds, ring-shaped sun, a flat mist stripe, identical blossoms — hand-made replacements.
- Storm front looked like a tornado; canopy cropped by the frame edge; a too-dark night shot; frozen shots.

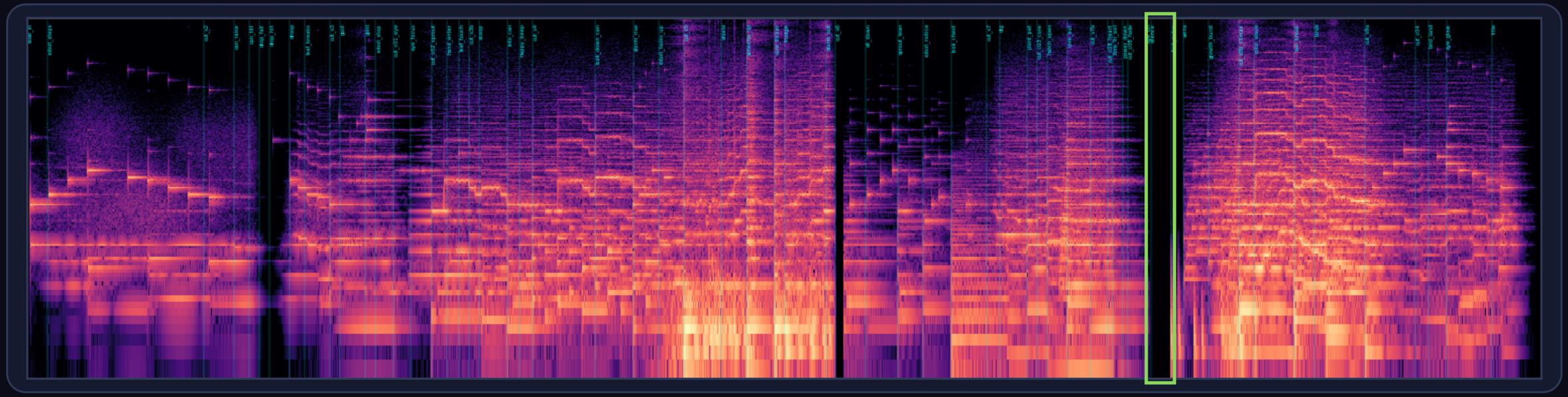
Audio

- Footsteps drifted off the feet (2.2 vs 2.0 steps/s) — derived from the animation's contact frames.
- The last drop's "tk" was masked by music — the score now rests for it.
- No hit on the heart flare at 41.2 s; the final note was cut off — both fixed.

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Sound, from nothing

Every instrument and every sound effect synthesized in numpy/scipy — no samples



True digital silence: the robot powers down

Score

- Two themes: the robot (music box) and life (ocarina), united at the climax with a key lift.
- Hand-built instruments: felt piano, string ensemble, choir, harp, taiko, celesta; convolution reverb.

Sound design

- Wind, storm, glass, birdsong, a stream; a tiny melodic robot voice that returns, softer, at the end.
- Event times read from the cue sheet and the shot files — retime one cue and audio follows.

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What made it work

Lessons for anyone directing a team of AI agents

1 Rules before talent

A shared engine, palette and contract let 11 agents work at once and still produce one coherent film.

2 Make the canon concrete

A pixel-exact model sheet beat any written description of the character.

3 Give agents eyes

Every agent rendered its own frames and looked at them, with at least four critique cycles.

4 One clock for everything

A single cue sheet drove picture, music and sound, so a retime is one edit.

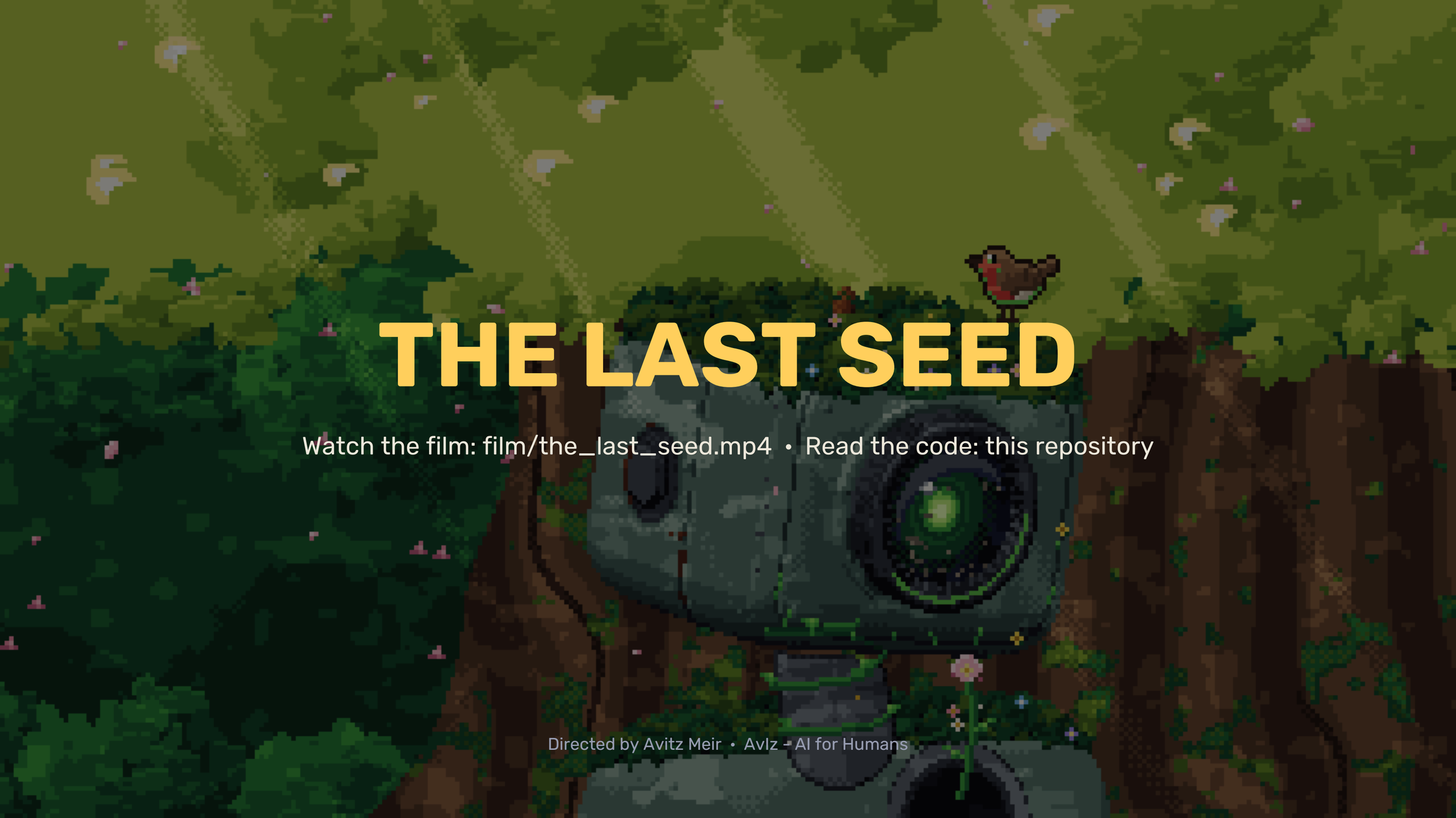
5 Hire a critic

The fresh-eyes reviewer found what the makers could not: a blank frame, a 4-pixel jitter, a hue jump.

6 Direct, don't micromanage

Clear intent and acceptance criteria per shot; owners fix their own work.

```
Run it yourself:  pip install numpy scipy pillow  ·  python render.py --sheet  ·  python render.py --final
```


THE LAST SEED
Watch the film: film/the_last_seed.mp4 • Read the code: this repository
Directed by Avitz Meir • Avlz - AI for Humans